

Project Design Phase

Solution Architecture

Date	15 February 2025
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Solution Architecture:

The Solution Architecture of the Stock Trading Platform defines the overall structure and interaction between system components developed using the MERN (MongoDB, Express.js, React.js, Node.js) stack. It provides a blueprint for transforming investor requirements into a secure, scalable, and user-friendly stock trading application.

The primary objective of the architecture is to enable users to perform stock trading activities such as account registration, authentication, stock purchasing, stock selling, portfolio management, and transaction tracking through a centralized platform.

The architecture consists of four major layers:

- 1. Presentation Layer (React Frontend)**
Provides an interactive user interface for registration, login, stock dashboard, portfolio management, and transaction history.
- 2. Application Layer (Node.js & Express.js)**
Handles business logic, REST API services, stock trading operations, portfolio calculations, and communication between frontend and database components.
- 3. Security Layer (JWT Authentication)**
Ensures secure access through token-based authentication and authorization mechanisms, protecting user accounts and transactions.
- 4. Data Layer (MongoDB Database)**
Stores user profiles, stock holdings, transaction records, portfolio information, and account balances.

The proposed architecture ensures modularity, scalability, maintainability, and secure data management. It enables seamless interaction between users and the stock trading system while supporting future enhancements such as real-time market data integration, advanced analytics, and mobile application support.

Example - Solution Architecture Diagram:

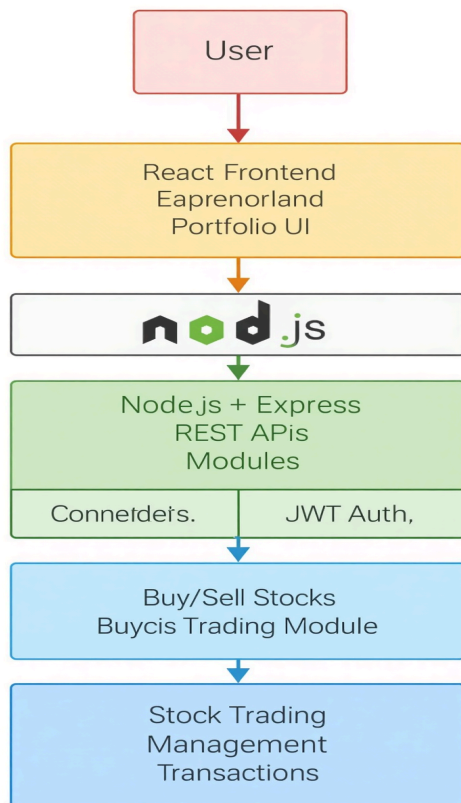


Figure 1: System Architecture and Data Flow of the Stock Trading Platform

Reference:

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>